

Trained Models

Model 1: risk_model_vitals_basic.pkl

Example input data:

```
{
  "Body Height": 165,
  "Body Weight": 60,
  "Body mass index (BMI) [Ratio]": 22,
  "Diastolic Blood Pressure": 80,
  "Systolic Blood Pressure": 118,
  "Respiratory rate": 16
}
```

Example Output Prediction: Patient condition appears stable.

Example input data:

```
{
  "Body Height": 170,
  "Body Weight": 75,
  "Body mass index (BMI) [Ratio]": 26,
  "Diastolic Blood Pressure": 65,
  "Systolic Blood Pressure": 135,
  "Respiratory rate": 28
}
```

Example Output Prediction

```
{
  High Risk Detected!
  Confidence: 78.00%
  Recommendation:
  Monitor respiratory function closely.
}
```

The model is a supervised machine learning classifier trained on structured clinical data. It predicts patient risk using six vital sign features—height, weight, body mass index (BMI), diastolic and systolic blood pressure, and respiratory rate—and produces both a binary risk classification and an associated probability score to support clinical decision-making. The model achieved an accuracy of 93% on the test dataset, demonstrating reliable performance in identifying high-risk patients.

Model 2: risk_model_v3_clinical.pkl & risk_model_v3_features.pkl

Example Input:

```
{
  "Body Height": 168,
  "Body Weight": 70,
  "Body mass index (BMI) [Ratio]": 24.8,
  "Heart rate": 115,
  "Body temperature": 38.7,
  "Pain severity - 0-10 verbal numeric rating [Score] - Reported": 6,
}
```

```
"Glucose [Mass/volume] in Blood": 145,  
"Hemoglobin [Mass/volume] in Blood": 11.5  
}
```

Example Output:

```
{  
    HIGH RISK DETECTED (Multi-Role Assessment)  
    Confidence: 94.00%  
  
    Doctor Assessment:  
    - Possible infection or inflammatory response.  
    - Elevated glucose level (possible hyperglycemia).  
    - Low hemoglobin (possible anemia).  
  
    Nursing Observations:  
    - Tachycardia observed.  
    - Patient experiencing moderate to high pain.  
  
    Lab Indicators:  
    - Blood glucose above normal range.  
    - Hemoglobin below normal range.  
  
    Immediate clinical evaluation recommended.  
}
```

The second model (Model V3) is a supervised machine learning classifier trained on structured clinical data derived from patient observations. It utilizes eight key input features, including physiological measurements (body height, body weight, body mass index), vital signs (heart rate, body temperature), patient-reported symptom severity (pain score), and laboratory indicators (blood glucose and hemoglobin levels). To ensure model validity, features used in defining the target variable were excluded during training to prevent data leakage. The model achieved an accuracy of approximately 93% on the test dataset, with high recall for high-risk cases, indicating strong capability in identifying patients requiring immediate clinical attention.